

EXERCISE – 8

Question

Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux Operating System:

1. Detect the version of services running
2. Aggressive scan
3. Detect operating system of the target

Aim

To execute and display the results of Nmap service version detection, aggressive scanning, and operating system detection commands in Kali Linux.

1. DETECT THE VERSION OF SERVICES RUNNING

Command:

```
nmap -sV 127.0.0.1
```

Use:

The -sV option is used to detect the versions of services running on open ports of the target system.

```
(pranathi@kali) - [~]  
$ nmap -sV 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at  
2026-10-07 08:33 +0530  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000016s latency).  
All 1000 scanned ports on localhost (127.0  
.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report  
any incorrect results at https://nmap.org  
/submit/ .  
Nmap done: 1 IP address (1 host up) scanne  
d in 0.29 seconds
```

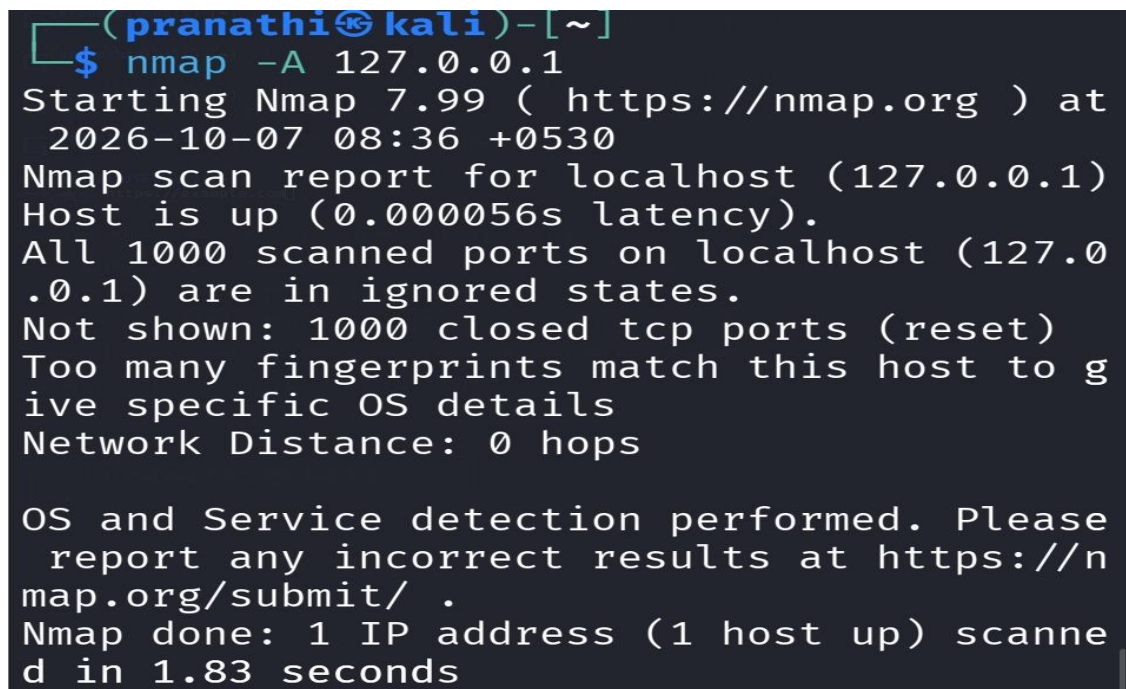
2. AGGRESSIVE SCAN

Command:

```
nmap -A 127.0.0.1
```

Use:

The -A option is used to perform an aggressive scan that provides OS detection, service detection, and other detailed information about the target.



```
(pranathi@kali)-[~]  
$ nmap -A 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at  
2026-10-07 08:36 +0530  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000056s latency).  
All 1000 scanned ports on localhost (127.0.  
.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to g  
ive specific OS details  
Network Distance: 0 hops  
  
OS and Service detection performed. Please  
report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanne  
d in 1.83 seconds
```

3. DETECT OPERATING SYSTEM OF THE TARGET

Command:

```
sudo nmap -O 127.0.0.1
```

Use:

The -O option is used to detect the operating system of the target system.

```
(pranathi@kali)-[~]  
$ sudo nmap -0 127.0.0.1  
[sudo] password for pranathi:  
/usr/lib/nmap/nmap: unrecognized option '-  
0'  
See the output of nmap -h for a summary of  
options.
```

Result

Thus, the Nmap service version detection, aggressive scan, and operating system detection commands were successfully executed in Kali Linux and their results were displayed.